Train a Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach reproducibility with the documented error intact.

Separate learned preparation from fixed conversion and preserve the optional cutoff-search topic with its own data and keys.

Keep one habit: Assess the new fitted state, unchanged predictions, input assumptions and bounded validation claim.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Connect **training**-only scaling and neighbour **prediction** in one **pipeline**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. Assemble the two previous ideas into a complete runnable program, while keeping hand tracing available without devices. Distinguish a **fitted state** from settings and a run record.

Require students to **reproduce** individual **prediction**s, including errors, rather than merely quote a score.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

It learns each feature minimum and range from **training**. Reuse the same minima, ranges, feature order and units to transform **training** and validation examples.

Printed lesson and workbook

Optional Python 3 editor with no packages required

Three cards labelled Fit **Predict** and Evaluate

Blank run record with split IDs feature order settings and **output**s

Apply the connected idea: Searching candidate rules

Teach two 45-minute sessions using fictional sensor readings and reviewed labels. This is a small supervised **fitting** algorithm with a supplied rule form, not a complete decision-tree implementation.

Separate the **prediction** boundary from the candidate replacement rule. The same split and baseline principles apply, and a zero **training error** must never be described as proof of generalization.

Readiness answer

It chooses its constant from **training** label counts, using a declared tie rule. Accuracy is then measured by matching the fixed **prediction** to validation labels.

Printed lesson and workbook

Four **training** cards and four validation cards with distance and reference label

Optional Python 3 editor with no extra packages

Blank table for cutoff **prediction**s errors and selected rule

Opening retrieval

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and rank **training** neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a majority label baseline using **training** labels only

Reference example, not the only acceptable wording: **Training** has four clear and two blocked, so the constant is clear. A tie chooses clear under the declared classroom rule.

Validation labels score the stored **prediction**; using their majority to choose it would change the **fitting** procedure.

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

Look for: Define the population and conditions in an intended-use statement

Reference example, not the only acceptable wording: Compare intended conditions with observed photo counts and distinct source plants. In Coverage v1, A has counts9,0,1,0 and B3,3,3,3 across day/A,day/B,evening/A,evening/B. Copying rows does not fill missing cells; coverage is not measured accuracy and unknown class/species selection remains a limit.

End goal

By the end, I can explain train a classifier using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Nearest Neighbors

Baselines First

Sampling and Intended Use

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Training: spots 0:H,2:H,4:S,6:S. Compare rules S if spots>=1,>=3,>=5. Choose by **training** accuracy, then **predict** held-out 1:H and 5:S without revising. Report both stages.

Concept to explain

Training fits model **parameter**s from labeled examples; **prediction** applies the frozen fitted model to new feature rows.

Term	Meaning
fit	learn parameters from training data
parameter	learned numeric model value
predict	apply fitted model to new features
pipeline	ordered preprocessing and model steps

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach fit, predict and evaluate as separate operations

> A unit conversion is fixed; this main min-max scale is fitted.

Complete main source and feature contract

A **pipeline** connects steps so a new query receives the same preparation as the **training** examples. This week combines the **training**-only scale introduced in Week 8 with a one-neighbour classifier. We will fit once, keep the **fitted state**, and call one **prediction** function for every raw query.

Use the same fictional trip data. Each feature pair is ordered (distance in cm, brightness in sensor units). **Training** IDs are A,B,C.

Validation IDs V1,V2,V3 belong to separate trips; related trips were grouped before splitting. Use min-max scaling, sum of absolute scaled differences, k=1 and alphabetical ID order for exact distance ties.

Training ID	Feature pair	Label
A	(10,200)	clear
B	(20,100)	blocked
C	(20,200)	blocked

The **input contract** requires two finite numeric features in the stated order and units, with no missing values. **Training** also needs unique IDs, known labels, at least one row and nonzero ranges.

These assumptions must be checked before this short classroom program is used; its explicit guard only checks zero ranges.

Trace saved state and printed outputs

fit returns minima (10,100), ranges (10,100) and scaled **training** examples. **predict** transforms a raw query with that state, ranks distances, then returns the selected ID and label. zip pairs matching features; sum adds their absolute differences. sorted uses distance first and unique ID second.

Run the whole script in Python 3. It prints ('A', 'clear'), ('C', 'blocked'), ('B', 'blocked') on separate lines. No validation labels enter fit or **predict**.

This is a small **pipeline** written with functions, not the scikit-learn **Pipeline** class.

Term	Meaning
Pipeline	An ordered sequence of data transformations and prediction steps
Fitted state	The learned values and stored examples needed to make predictions
Reproduce	Obtain the same result using the same recorded inputs and procedure
Input contract	The documented feature names order units and accepted values

Trace raw query, state reuse and swapped meanings

Two stages with one fitted state

STAGE	INPUT AND ACTION	RESULT
Fit once	Training rows only	Scale plus stored rows
Predict each query	Raw pair plus state	Neighbour ID and label
Evaluate afterward	Prediction plus reference	Recorded match or error

Reference labels are used after prediction to score results. They do not enter the fitted scale or stored neighbour pool.

For raw query (11,120), scale gives (0.1,0.2). Distances to A,B,C are 0.9,1.1,1.7, so the function returns A and clear. It does not recalculate minima or add the query to stored rows.

The returned ID is useful for checking the reasoning, but it is not **evidence** that the label is true.

> **Trace (11,120) through both functions. What does fit store, what does predict reuse, and at which point may a validation reference label be compared?**

Store the **fitted state** together with the **input contract** and method settings. A model without its scale or feature order is incomplete.

Swapping the raw pair to (120,11) produces B and blocked here; the function cannot detect that the caller swapped the meanings.

Read separate unit-conversion facts

Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.

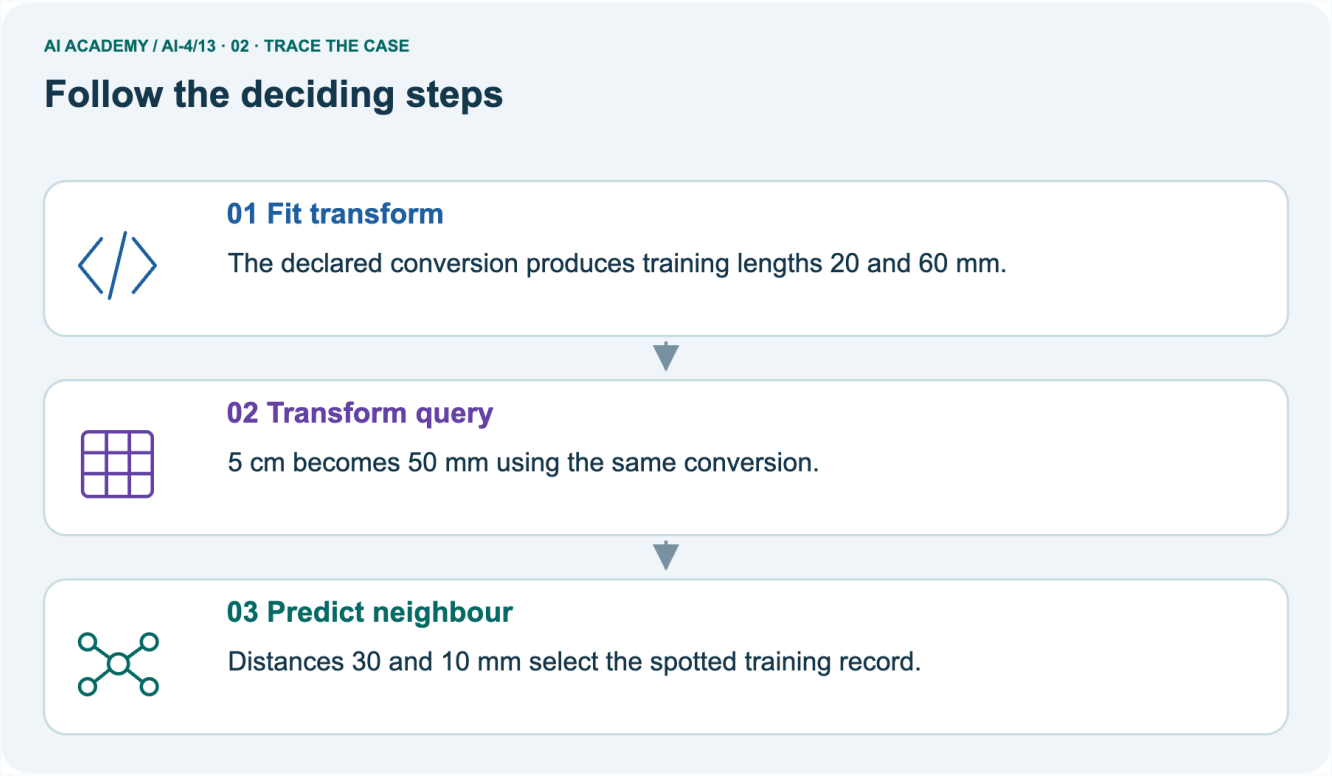
Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance. Query is 5 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The main pipeline learns minima and ranges from training. The pictured cm-to-mm conversion is different: multiplying by10 is a declared unit conversion, not a parameter learned from examples. Training lengths2/6cm become20/60mm; query5cm becomes50mm and selects the spotted record at distance10mm. Comparing the unconverted numeric5with20/60mixes units and incorrectly changes the neighbour.

Record whether each preparation step is a fixed rule or learned state; both must be applied consistently.

Trace consistent units and the changed defect



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit transform	The declared conversion produces training lengths 20 and 60 mm.
2. Transform query	5 cm becomes 50 mm using the same conversion.
3. Predict neighbour	Distances 30 and 10 mm select the spotted training record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Distances 30 and 10 mm select the spotted **training** record. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

A reproducible toy **pipeline** can still perform poorly; evaluation must use separate records and appropriate group splits.

Change: Transform **training** but leave query numerically at 5.

Prediction: Distances become 15 and 55, incorrectly selecting healthy.

Reason: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

▼ Optional cutoff-search contract and keys

The majority baseline from Week 10 ignored scene features. This week a small **fitting** algorithm will choose a rule that uses one feature. We supply the rule form: **predict** blocked when a distance reading is at or below a cutoff, otherwise **predict** clear. The algorithm will try three cutoffs and keep the one making the fewest **training** mistakes.

The teacher-authored fictional rows contain a sensor distance in centimetres and a separate reviewed lane label, blocked or clear. The reading is available before **prediction**; the label comes from the scene review.

A single distance need not describe every obstruction. These numbers are for learning, not real robot control. Each row comes from a different trip, and **training** and validation trips are separate.

Before scoring, declare cutoffs 3, 5 and 7 cm in ascending order, **predict** blocked at equality, and keep the smaller cutoff if **training error**s tie. This search tests only these three rules with this direction.

It does not find the best possible model or prove that distance causes the label.

The **output** lines are 3 1, 5 0, 7 1, then chosen 5. Cutoff 3 misses the blocked row at 4 cm. Cutoff 5 matches all four **training** labels. Cutoff 7 wrongly marks the 6 cm clear row as blocked.

Fitting stores cutoff 5; **prediction** then applies that stored rule to a new distance.

best_errors starts above any possible mistake count. The strict < keeps the first tied candidate; ascending order makes that the smaller cutoff.

The code assumes nonempty **training** data, numeric distances and the two allowed labels. Validate those assumptions before using another dataset.

Guided: at 5 cm the selected rule **predict**s blocked because equality is included. The rule misses validation rows 2 and 4; baseline clear misses rows 1 and 4. Both score 50 percent, a zero percentage-point difference.

Exit: cutoff 5 has zero mistakes on four **training** rows among the three supplied candidates, but scores 2/4 on validation and ties the baseline. This small comparison does not prove broad performance or global optimality.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Run a starter classifier and record data version, split, settings, and **prediction**s.

MODEL IDEA **Training** fits model **parameter**s from labeled examples; **prediction** applies the frozen fitted model to new feature rows.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Load data version v1 and a saved train/test split. Fit pre**process**ing on **training** rows only, then fit the classifier.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Load data version v1 and a saved train/test split.

STEP 2 Fit pre**process**ing on **training** rows only, then fit the classifier.

STEP 3 Freeze the **pipeline** and **predict** twenty test rows once.

STEP 4 Save settings, **prediction**s, and metrics so the experiment can be **reproduce**d.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Training is (0,H),(1,H),(4,S),(5,S). **Candidate rule predict**s S for spots>=3. Held-out cases are (2,H),(3,S).

Apply the rule and distinguish **fitting** from inference.

Exact explanation to model

Predictions H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

Explain the added worked example

Key terms

Apply the connected idea: Searching candidate rules

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Train a Classifier

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and **predict** the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Give the IDs and labels printed for V1,V2,V3. Use references clear,blocked,clear to calculate accuracy and the **training**-majority baseline. Which error should repeat in an unchanged rerun?

Teacher check: V1 selects A clear, V2 C blocked, V3 B blocked. Accuracy is 2/3. The **training**-majority blocked baseline is correct only for V2, scoring 1/3.

V3 is the repeatable error; repeated **output** does not turn it into a correct **prediction**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for A reproducible classifier **pipeline**. Proposed explanation: “A **pipeline** prevents every leak automatically”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Correct sequencing cannot fix a fit call that receives held-out rows or features unavailable at **prediction** time.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Record **training input**s/labels, unit conversion, k=1, distance rule and query transformation; correct result is spotted from distance10 mm. Changed-case check: Distances become 15 and 55, incorrectly selecting healthy.

The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Changed-case reasoning: Distances become 15 and 55, incorrectly selecting healthy. The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: **Predict** first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: **Training** is (0,H),(1,H),(4,S),(5,S). **Candidate rule predict**s S for spots>=3. Held-out cases are (2,H),(3,S).

Write the question to explain: Apply the rule and distinguish **fitting** from inference.

Reveal after **prediction**s: **Prediction**s H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

Use the earlier transfer question as additional practice. Reserve the selected fresh P/Q/R fitted-pipeline task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: fit stores minima (10,100), ranges (10,100) and A (0,1) clear, B (1,0) blocked, C (1,1) blocked. **predict** transforms (11,120) to (0.1,0.2), measures 0.9,1.1,1.7 and returns A clear. Compare the reference only afterward.

Exit: reproducible leakage is still leakage. Restore the split, fit on **training** only, use the frozen state for every raw validation query, then score and record all **output**s.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve training-only scaling and the stored minima and ranges. State the input contract.
7 to 18 minutes	Use the three cards to place data and labels at the correct stage. Complete W1.

Stage	Teacher action
18 to 32 minutes	Run or trace the complete script. Unpack fitted state and explain zip sum and sorted.
32 to 45 minutes	Use the diagram and guided question, then complete W2 from the printed outputs.
Session 2 start to 8 minutes	Retrieve why predict must reuse the fitted state and receive raw feature pairs.
8 to 20 minutes	Complete W3 and compare two run records. Discuss why equal scores can hide different predictions.
20 to 35 minutes	Collect W4 independently before support. Require transformed queries and ID-linked predictions.
35 to 45 minutes	Use W5 and the exit question to audit misuse. Offer the fresh retry after feedback and record support.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve training-majority fitting and validation scoring. Establish the distinct-trip fictional data and distance unit.
7 to 18 minutes	State candidates, direction, equality and tie policy before counting. Predict all four training rows for cutoff 3 by hand.
18 to 32 minutes	Trace the loop and complete W1 and W2. Name the stored parameter and distinguish fitting from later prediction.
32 to 45 minutes	Reveal validation and compute both methods on the same rows. Complete the guided boundary and error-pattern question.
Session 2 start to 8 minutes	Retrieve the difference between the prediction <= and search < comparisons.
8 to 20 minutes	Complete W3 and inspect both bug examples. Explain that a tie policy is a convention, not evidence of future superiority.
20 to 35 minutes	Collect W4 independently before hints. Require every candidate count, selected cutoff, predictions and baseline comparison.
35 to 45 minutes	Use W5 and the exit claim. Give the fresh retry after feedback and record support.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

▼ Extension teaching cycles

Connected teaching cycle: Searching candidate rules

Target: Fit a **threshold** rule by comparing **training error**s

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: List errors for cutoffs 3, 5 and 7 on the worked **training** rows. Identify the mistaken row for each imperfect candidate. Which cutoff is stored?

Teacher check: Errors are 1,0,1. Cutoff 3 misses row 2 at 4 cm; cutoff 7 misses row 3 at 6 cm. Cutoff 5 has no **training** mistakes and is stored as the best of these three candidates.

Individual check: For Searching **candidate rule**s, use its worked case to evaluate this proposed explanation: “Zero **training error** means perfect **prediction**s”. Explain the deciding step.

Expected correction: The rule was selected using those labels. New examples can disagree, as the two validation mistakes show.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Main pipeline2/3 versus baseline1/3; save input contract and complete fitted state. Tie(10,100) chooses A; outside-range(12,80) is allowed under formula; missing violates input contract; swapped features are undetected by this short program; constant training feature triggers guard.

Selected independent minima(0,20),ranges(20,40),queries(0.1,0.2),(0.9,0.9),(0.2,-0.2),IDs P/R/Q,labels blocked/clear/clear,2/3 versus clear baseline1/3. Optional cutoff search3/5/7 errors1/0/1,choose5;validation2/4 ties clear baseline2/4. No final-test result.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Starter Classifier Run: trace code, **predict output**s, execute the fixed **pipeline**, and compare recorded results with a peer.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A **pipeline** prevents every leak automatically: Correct sequencing cannot fix a fit call that receives held-out rows or features unavailable at **prediction** time.

A seed is the whole reproducibility record: This program is deterministic. Data code split feature order settings and runtime still matter; a seed cannot repair changed **input**s.

Same score means same **prediction**s: Compare **prediction**s by example ID. Different error patterns can produce the same accuracy.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

Zero **training error** means perfect **prediction**s: The rule was selected using those labels. New examples can disagree, as the two validation mistakes show.

Every comparison should include equality: **Prediction** and candidate replacement serve different purposes. Follow the specified boundary and tie policy separately.

The best tested rule is the best possible model: The search covers only the declared candidates and rule form, evaluated on a small set. Other forms and future cases remain unknown.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh P/Q/R fitted-pipeline task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR The model keeps learning whenever it sees test examples during evaluation.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new **training** rows. Fit the same **pipeline**. **Predict** validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked.

Give fitted minima and ranges, transformed queries, selected IDs, **prediction**s, accuracy and **training**-majority baseline.

Minima (0,20), ranges (20,40). Scaled queries are (0.1,0.2),(0.9,0.9),(0.2,-0.2). Selected IDs P,R,Q give blocked,clear,clear. Accuracy is 2/3; the **training**-majority clear baseline gets 1/3. The third query remains wrong.

Validation rows do not change the **fitted state**.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Separate fit predict and evaluate

List what fit learns or stores, what **predict** takes as **input**, and what evaluation compares. State where the validation reference labels belong.

fit learns **training** minima and ranges and stores scaled **training** rows with IDs and labels. **predict** takes a raw feature pair and the **fitted state**. Evaluation compares a **prediction** with its reference label afterward.

Validation references belong only in evaluation, not in **fitting** or the neighbour pool.

W2 Trace and score the saved run

Give the IDs and labels printed for V1,V2,V3. Use references clear,blocked,clear to calculate accuracy and the **training**-majority baseline. Which error should repeat in an unchanged rerun?

V1 selects A clear, V2 C blocked, V3 B blocked. Accuracy is 2/3. The **training**-majority blocked baseline is correct only for V2, scoring 1/3.

V3 is the repeatable error; repeated **output** does not turn it into a correct **prediction**.

W3 Repair an incomplete run record

A classmate saves only k=1 and a score of 2/3. Name the information needed to **reproduce** this run. Explain why the same score alone is insufficient.

Record the data and code versions, exact split IDs, feature names order and units, fitted minima and ranges, stored **training** rows, distance and tie policy, zero-range policy, runtime version and **prediction**s linked to validation IDs.

A score hides which examples were correct; different **prediction**s can have equal accuracy.

W5 Diagnose pipeline misuse

Explain and repair three mistakes: **predict** receives an already scaled query; fit receives **training** plus validation rows; a query uses brightness before distance. Does repeating each mistaken run make it valid?

predict expects raw features, so a pre-scaled query is scaled again; pass the raw pair. **Fitting** on validation leaks evaluation information; refit on **training** only and reevaluate.

Swapped features violate the **input contract**; restore distance then brightness in the stated units. Repeating a mistake shows repeatability, not validity.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Predictions H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

Additional transfer practice question

A classifier fits on rows 1–80 and is evaluated on those same rows. Its accuracy is 100%. What additional **evidence** is needed before claiming generalization?

Reasoned practice answer

Evaluate on independent examples not used to fit or tune the model, with appropriate split discipline. **Training** accuracy alone measures fit to seen data.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact

Criterion	0 points	1 point	2 points
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification **threshold**.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Perfect **training** accuracy proves perfect performance on new examples.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Threshold 1 gets 3/4, 3 gets 4/4, 5 gets 3/4. Select 3; held-out **prediction**s are H,S, giving 2/2. This finite search is a simple **training** procedure. Tiny perfect test performance does not justify broad claims.

Explanations for the additional workbook examples

W1 Separate fit predict and evaluate

W2 Trace and score the saved run

W3 Repair an incomplete run record

W4 Build and trace a fresh fitted state

Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new **training** rows. Fit the same **pipeline**. **Predict** validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked.

Give fitted minima and ranges, transformed queries, selected IDs, **prediction**s, accuracy and **training**-majority baseline.

Minima (0,20), ranges (20,40). Scaled queries are (0.1,0.2),(0.9,0.9),(0.2,-0.2). Selected IDs P,R,Q give blocked,clear,clear. Accuracy is 2/3; the **training**-majority clear baseline gets 1/3. The third query remains wrong.

Validation rows do not change the **fitted state**.

W5 Diagnose pipeline misuse

Assess independent transfer

Selected case answer

Feedback that the learner uses

▼ Extension questions and checked keys

Searching candidate rules — W1 State the search before running it

Write the rule direction, equality behavior, candidate cutoffs and tie policy. Identify what humans supplied and what **fitting** selects.

Predict blocked for distance <= cutoff and clear otherwise. Try 3,5,7 cm; choose the smaller cutoff on tied **training error**s.

Humans supplied the feature, rule form, candidates and policy; **fitting** selects the cutoff using **training** labels.

Searching candidate rules — W2 Trace the training mistakes

List errors for cutoffs 3, 5 and 7 on the worked **training** rows. Identify the mistaken row for each imperfect candidate. Which cutoff is stored?

Errors are 1,0,1. Cutoff 3 misses row 2 at 4 cm; cutoff 7 misses row 3 at 6 cm. Cutoff 5 has no **training** mistakes and is stored as the best of these three candidates.

Searching candidate rules — W3 Audit the boundary and comparison

What happens at distance 5 if **prediction** uses < instead of <= with cutoff 5? Explain why the search uses a strict < for error counts even though **prediction** uses <= for distance.

The changed **prediction** would be clear instead of blocked at 5 cm. **Prediction** <= includes equality in blocked.

Search < replaces a candidate only for fewer mistakes, preserving the first tied candidate; with ascending candidates that implements the smaller-cutoff policy.

Searching candidate rules — W4 Search and evaluate independently

New **training** rows are (1,blocked), (3,clear), (5,blocked), (7,clear). Try cutoffs 2,4,6, using the same direction and smaller-cutoff tie policy. Give every **training error** count and select a cutoff.

Validation rows are (2,blocked), (4,blocked), (6,clear), (8,clear). List the selected rule's **prediction**s and score. Compare with the **training**-majority baseline using clear on ties.

Training errors for 2,4,6 are 1,2,1. Choose 2 under the declared tie policy. Validation **prediction**s are blocked,clear,clear,clear, matching rows 1,3,4: 3/4 or 75 percent.

Training labels tie, so baseline clear matches rows 3,4: 2/4 or 50 percent. The selected rule gains one correct example or 25 percentage points on this small set only.

Searching candidate rules — W5 Limit the claim

The worked rule fits **training** perfectly but ties the baseline on validation accuracy. Explain why neither result proves it is the best possible model. Name two details to keep in the experiment record.

Perfect **training** fit describes the examples used for selection; the four-row validation score is only 50 percent and ties the baseline. Only three candidates of one form were searched.

Keep the candidates, all error counts, tie policy, chosen cutoff, split and row-aligned validation **prediction**s; none proves general superiority.

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Train Plant Classifier v1 without opening the final test labels during model choice; save the experiment record.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Label each tuple field and use one query at a time. Provide a blank fit/**predict**/evaluate diagram without answers. Extension: use query (10,100) in the worked model.

A and B both have scaled distance 1, C has 2; alphabetical ID order selects A clear. Explain why this convention must be recorded.

Require W4 minima (0,20), ranges (20,40), scaled queries (0.1,0.2),(0.9,0.9),(0.2,-0.2), IDs P,R,Q, **prediction**s blocked,clear,clear and scores 2/3 versus 1/3. Fresh retry: L (5,80) clear, M (15,40) blocked, N (15,80) blocked.

Fit gives minima (5,40), ranges (10,40). Queries (6,48),(14,76) scale to (0.1,0.2),(0.9,0.9), selecting L clear and N blocked. Require the **input contract** and a statement that **prediction** does not refit.

Delayed return

Suggested return points in this level: ai-4/14, ai-4/16, ai-4/19. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	

Record	Teacher evidence
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 14: Scores and Calibration Intuition. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 14: Scores and Calibration Intuition. Keep the ID-linked references, **prediction**s, fitted state and method settings. Confusion matrices follow in Week 15. Primary reference: https://scikit-learn.org/stable/common_pitfalls.html (consistent pre**process**ing and **training**-only **fitting**).

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

▼ **Extension support and reassessment**

Readiness and reassessment for Searching candidate rules

Use a number line and move a cutoff marker; place equality on the blocked side. Give a blank **prediction** grid but no error counts.

Extension: reverse tied candidate order in the supplied code and explain why the retained cutoff changes unless the tie policy is implemented explicitly.

Require W4 **training error**s 1,2,1, cutoff 2, validation 3/4 and baseline 2/4 with a 25 percentage-point difference. Fresh retry: **training** (2,blocked),(4,clear),(6,blocked),(8,clear), candidates 3,5,7.

Errors are 1,2,1; select 3. Validation (3,clear),(5,blocked),(7,clear),(9,clear) gives rule blocked,clear,clear,clear: 2/4. Baseline clear gives 3/4, so the rule is one correct example or 25 percentage points worse.

Require correct equality handling and a restrained claim.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Optional threshold lab · Teaching guidance**

REFERENCE · SECTION 7

Optional threshold lab · Teaching guidance

> Use with the separately named executable threshold lab, after the main pipeline route.

Complete retained lab guidance

Fit and test a threshold classifier

The program chooses a **threshold** using **training** labels only. Spotted is 1 and healthy is 0. Candidate **threshold**s lie between consecutive **training** values. Test rows are used after selection.

```
"""Fit a threshold using training data, then evaluate untouched examples."""
train = [(0, 0), (1, 0), (4, 1), (5, 1)]
values = sorted({x for x, y in train})
if len(values) < 2:
    raise ValueError("Two distinct training values are required")
candidates = [(a + b) / 2 for a, b in zip(values, values[1:])]


def errors(threshold):
    return sum(int(x >= threshold) != y for x, y in train)


threshold = min(candidates, key=errors)
test = [(2, 0), (3, 1)]
predictions = [int(x >= threshold) for x, y in test]
```

```
correct = sum(pred == y for pred, (x, y) in zip(predictions, test))
print(threshold, predictions, correct, len(test))
```

Expected **output**: 2.5 [0, 1] 2 2

Candidate **threshold**s 0.5, 2.5 and 4.5 have **training error**s 1, 0 and 1. The selected **threshold** is 2.5. Both test **prediction**s are correct.

This tiny artificial example demonstrates **fitting** and evaluation; it does not establish reliability on real leaf photos.

Independent code change

Change the final test row to (3, 0), keeping **training** unchanged. **Predict** before running.

Teacher answer: **Threshold** remains 2.5 and **prediction**s remain [0, 1], but only 1/2 test **prediction**s is correct.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 8

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Connect **training**-only scaling and neighbour **prediction** in one **pipeline**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min

> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Give the IDs and labels printed for V1,V2,V3. Use references clear,blocked,clear to calculate accuracy and the **training**-majority baseline.

Which error should repeat in an unchanged rerun?

Reasoned practice answer: V1 selects A clear, V2 C blocked, V3 B blocked. Accuracy is 2/3. The **training**-majority blocked baseline is correct only for V2, scoring 1/3.

V3 is the repeatable error; repeated **output** does not turn it into a correct **prediction**.

Check every learner before discussion: Use the worked case for A reproducible classifier **pipeline**. Proposed explanation: “A **pipeline** prevents every leak automatically”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Correct sequencing cannot fix a fit call that receives held-out rows or features unavailable at **prediction** time.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Transform **training** but leave query numerically at 5. Require a **prediction** and a reason before comparing. Expected result: Distances become 15 and 55, incorrectly selecting healthy.

Reason: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

Ask students to **predict** before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

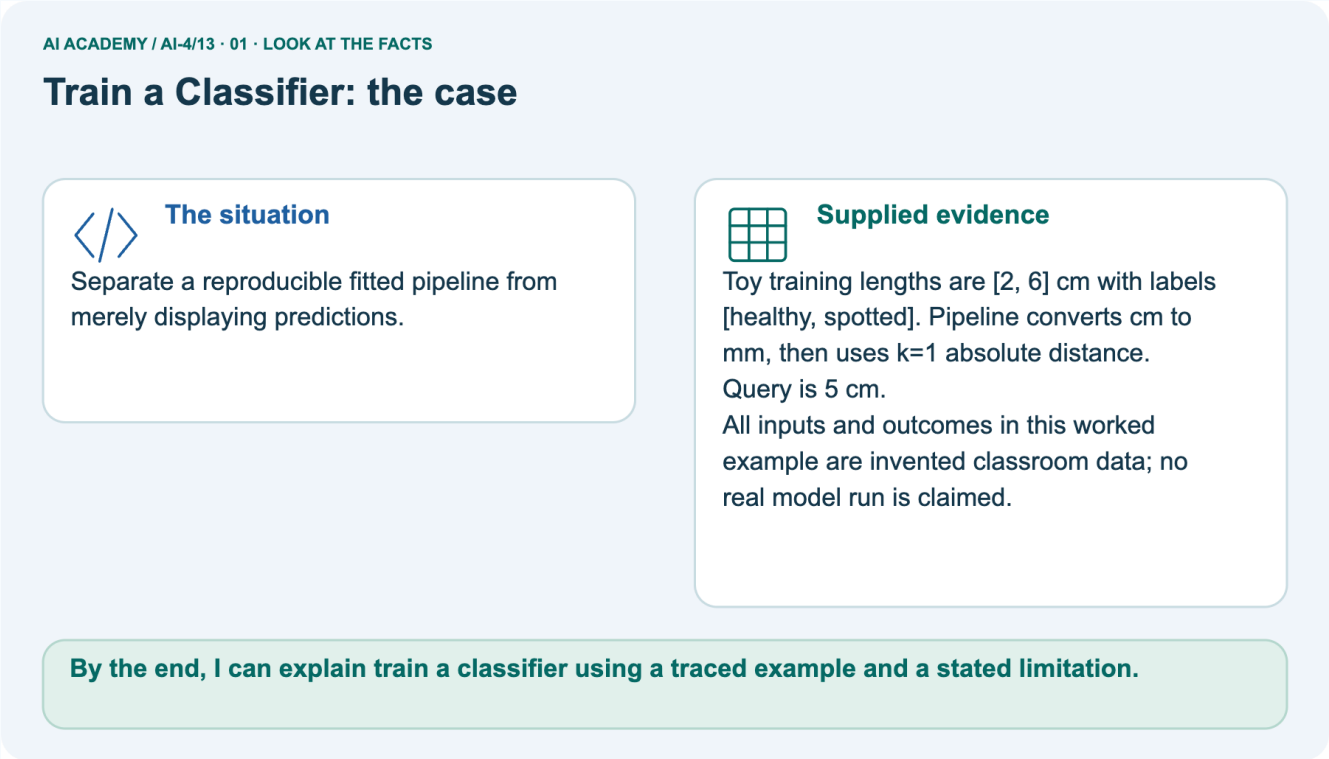
▼ Reference · Teach with the six pictures

REFERENCE · SECTION 10

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Train a Classifier: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

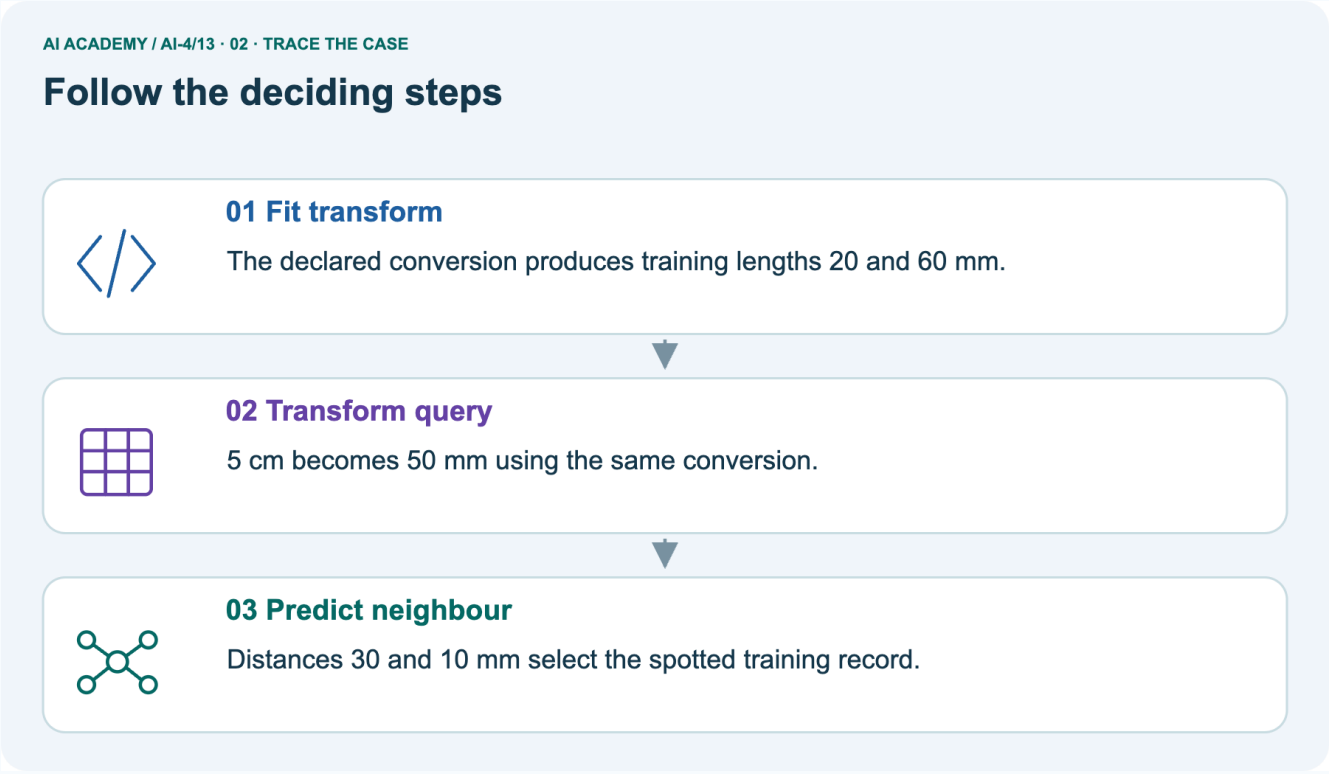
Separate a reproducible fitted **pipeline** from merely displaying **prediction**s. Supplied facts: Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance.

Query is 5 cm. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



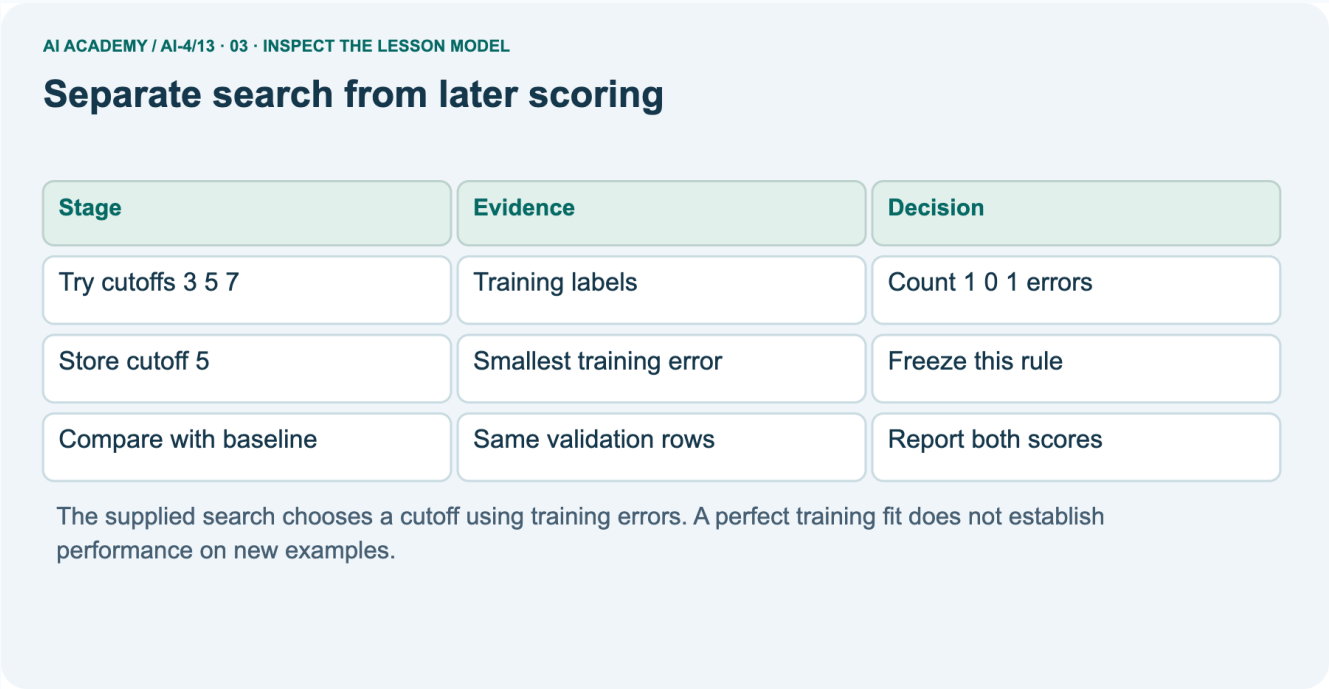
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit transform: The declared conversion produces **training** lengths 20 and 60 mm. → Transform query: 5 cm becomes 50 mm using the same conversion. → **Predict** neighbour: Distances 30 and 10 mm select the spotted **training** record.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Separate search from later scoring



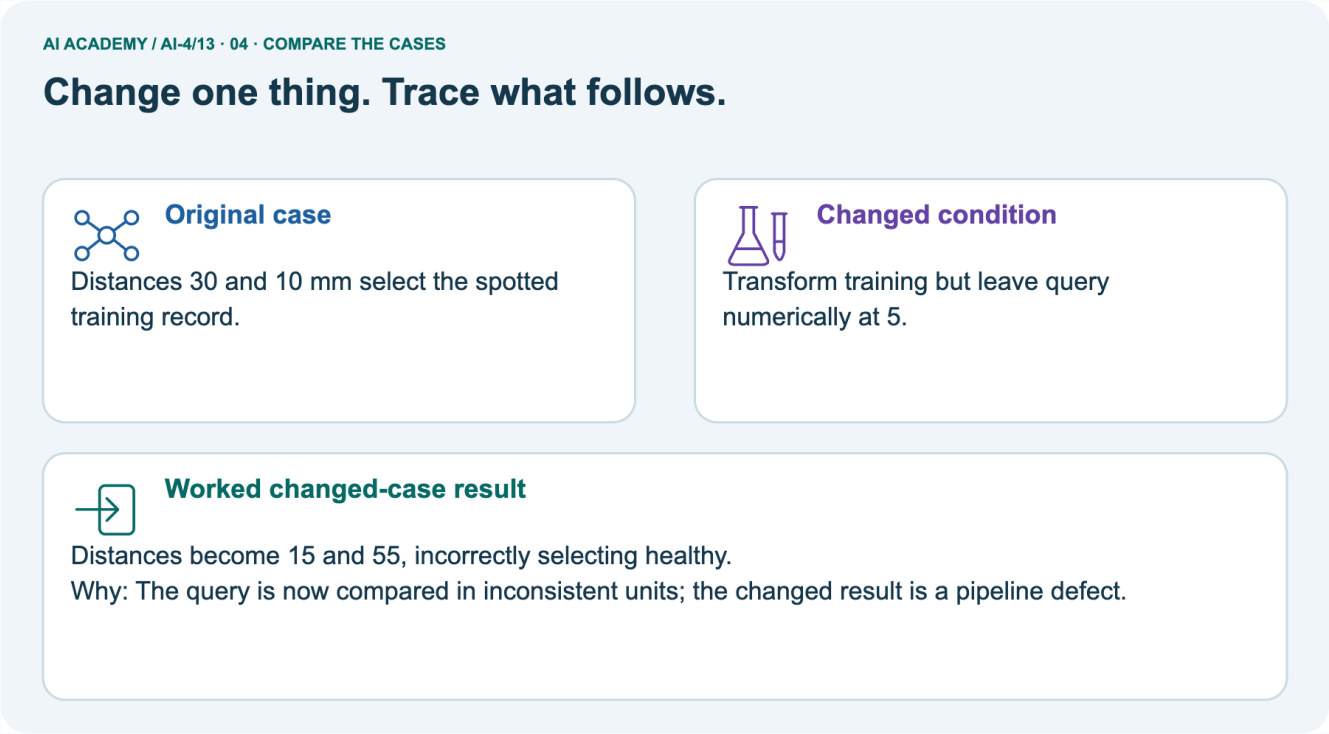
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate search from later scoring. Try cutoffs 3 5 7 | **Training** labels | Count 1 0 1 errors. Store cutoff 5 | Smallest **training error** | Freeze this rule. Compare with baseline | Same validation rows | Report both scores

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Distances 30 and 10 mm select the spotted **training** record. Changed condition: Transform **training** but leave query numerically at 5. Expected result: Distances become 15 and 55, incorrectly selecting healthy.

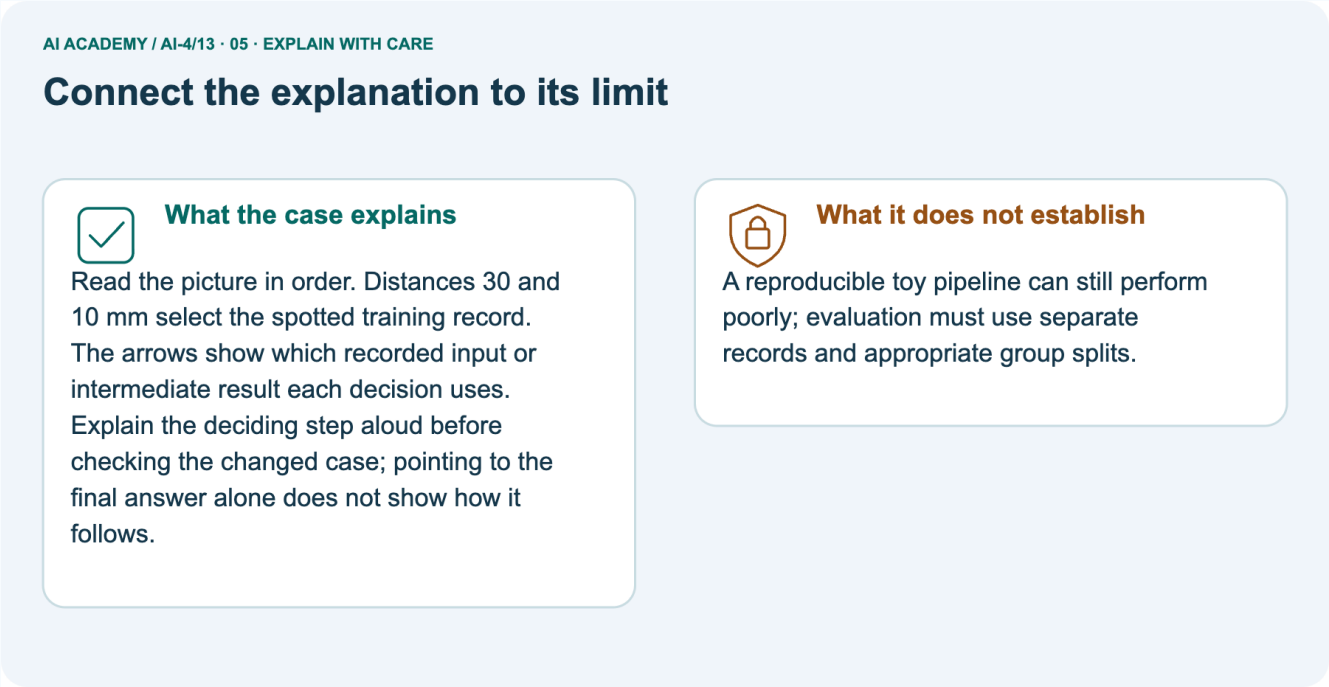
Reason: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: **Predict** before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Distances 30 and 10 mm select the spotted **training** record. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

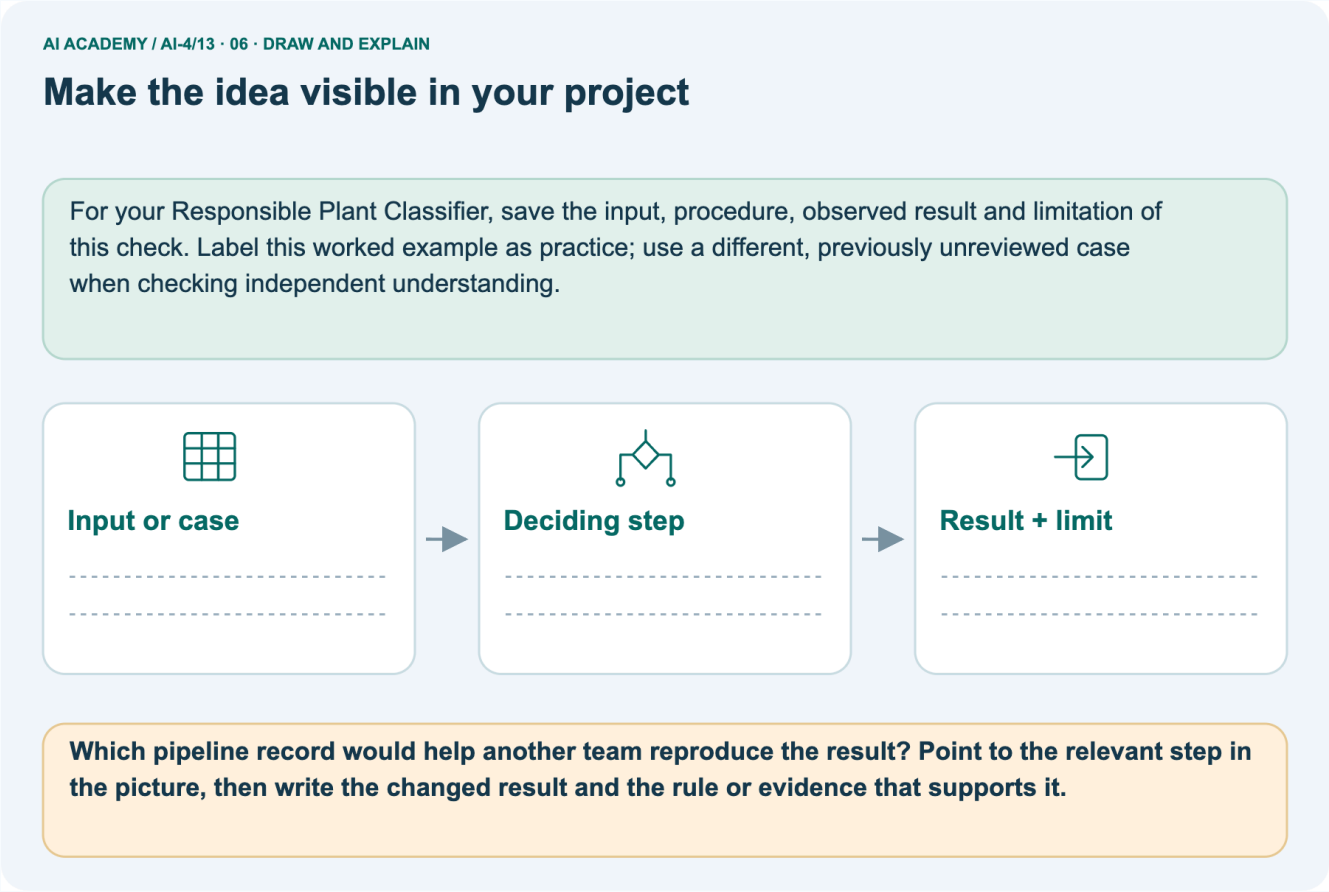
Limit: A reproducible toy **pipeline** can still perform poorly; evaluation must use separate records and appropriate group splits.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A reproducible toy **pipeline** can still perform poorly; evaluation must use separate records and appropriate group splits.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Record **training input**s/labels, unit conversion, k=1, distance rule and query transformation; correct result is spotted from distance10 mm.

Changed-case check: Distances become 15 and 55, incorrectly selecting healthy. The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Train a Classifier: the case

02. Follow the deciding steps

03. Separate search from later scoring

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.